

CODE SPORT



In this month's column, we discuss a medley of topics including solving cognitive intelligence puzzles, and how Python is getting used widely in AI and Natural Language Processing.

For the past few months, we have been discussing information retrieval and Natural Language Processing (NLP), along with the algorithms associated with them. In this month's column, we take a break from that discussion to look at some of the questions I have been getting from our readers recently. Given that we generally discuss coding problems, some readers have asked me whether I can discuss some puzzles. So instead of discussing math puzzles, I thought it may be interesting to try out something that tests our cognitive intelligence capabilities. Here is the first puzzle, which initially appears quite simple, but many people, including college graduates, seem to get it wrong on their first try.



There are three people — Andrew, Anne, and Bobby. Andrew is looking at Anne.

Anne is looking at Bobby. Andrew is married but Bobby is not married. Now the question to you is: Is a married person looking at an unmarried person? Choose the correct answer from the following options:

(A) Yes (B) No (C) Cannot be determined

Most people tend to choose the answer (C). Do you think the answer is correct? If not, why not? First take a couple of minutes to think through your reply. Well, the correct answer is (A). The key point to note is that each person can be in either a 'married' or 'unmarried' state. We know that Andrew is married and that Bobby is not. The reason most folks choose (C) is because there is no information regarding the marital state of Anne. But remember that the question is not whether Anne is married or not. The actual question is whether a married person is looking at an unmarried person. Anne can either be married or unmarried. If she

is married, then answer (A) is indeed a married person looking at an unmarried one since Bobby is unmarried. If Anne is unmarried, then too, (A) emerges as the correct answer since Andrew who is married, is looking at Anne who is unmarried. Hence the answer is (A) and not (C). The reason most people choose (C) is because they are poor in disjunctive reasoning, which requires all the different possibilities or outcomes associated with a given situation to be taken into account before arriving at a conclusion.

Here is another puzzle to exercise your brain. This being a simple puzzle, I would like you to answer this question very quickly, in your mind and without using pen and paper.



A bat and a ball cost \$1.10. The bat costs one dollar more than the ball. How much does the ball cost?

The immediate answer that most people give is 10 cents. However, if you work it out with a bit more careful thought, if the ball costs 10 cents, this makes the cost of the bat as one dollar and 10 cents since the bat costs one dollar more than the ball. The total cost of both bat and ball would then be \$1.20, which is incorrect since the total given in the puzzle is \$1.10. With a little bit of careful thought, you will find that the cost of the ball is five cents and that of the bat is one dollar five cents, which makes the total \$1.10. This problem is very simple as far as mathematics is concerned. But when asked to name the answer immediately, somehow most folks blurt out the answer as ten cents instead of as five cents. The reason for this cognitive dysfunction is because our quick thinking brain is deceived into coming up with an obvious answer. However, a little more careful thinking prevents this error of reasoning.